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# The Munro Bag

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# 1. Overview

## Intended User

Outdoor enthusiasts who are interested in bagging Munros in Scotland. Beginners or experienced hikers who want a better way to organise and track their progress.

## Problems Addressed

There are 282 Munros in Scotland which can be difficult for hikers to keep track of. This application will allow users to keep track of completed and remaining Munros in a centralized location.

## Value Added

The application will allow users to quickly search their progress of completed or remaining Munros. Users will be able to upload photos assigned to each Munros and add personal notes. This will make it easier for users to plan future hikes.

# 2. User Stories

## MUST

### M1 – User Authentication

As a user, I want to register and log in to the application, so that my personal account data and climbing records are securely stored and persist across sessions.

### M2 – View Munro Data (Database-backed model)

As a user, I want to view a list of Munros retrieved from a database, with key information (height, region, difficulty), so that I can plan which Munros I want to climb.

### M3 – Record a Completed Munro (User input saved and reused)

As a user, I want to record a completed climb for a Munro, including the date, completion time, rating, and comments, so that I can keep a personal log of my climbing experiences.

### M4 – Track Personal Progress

As a user, I want to see which Munros I have already completed clearly highlighted across the application, so that I can easily track my overall progress.

## SHOULD

### S1 – Search Functionality

As a user, I want to search for a specific Munro by name or region, so that I can quickly find relevant information.

### S2 – User Profile and Statistics

As a user, I want to view a personal profile showing derived statistics, such as total Munros completed and favourite Munro, so that I can reflect on my climbing history.

## COULD

### C1 – Photo Uploads

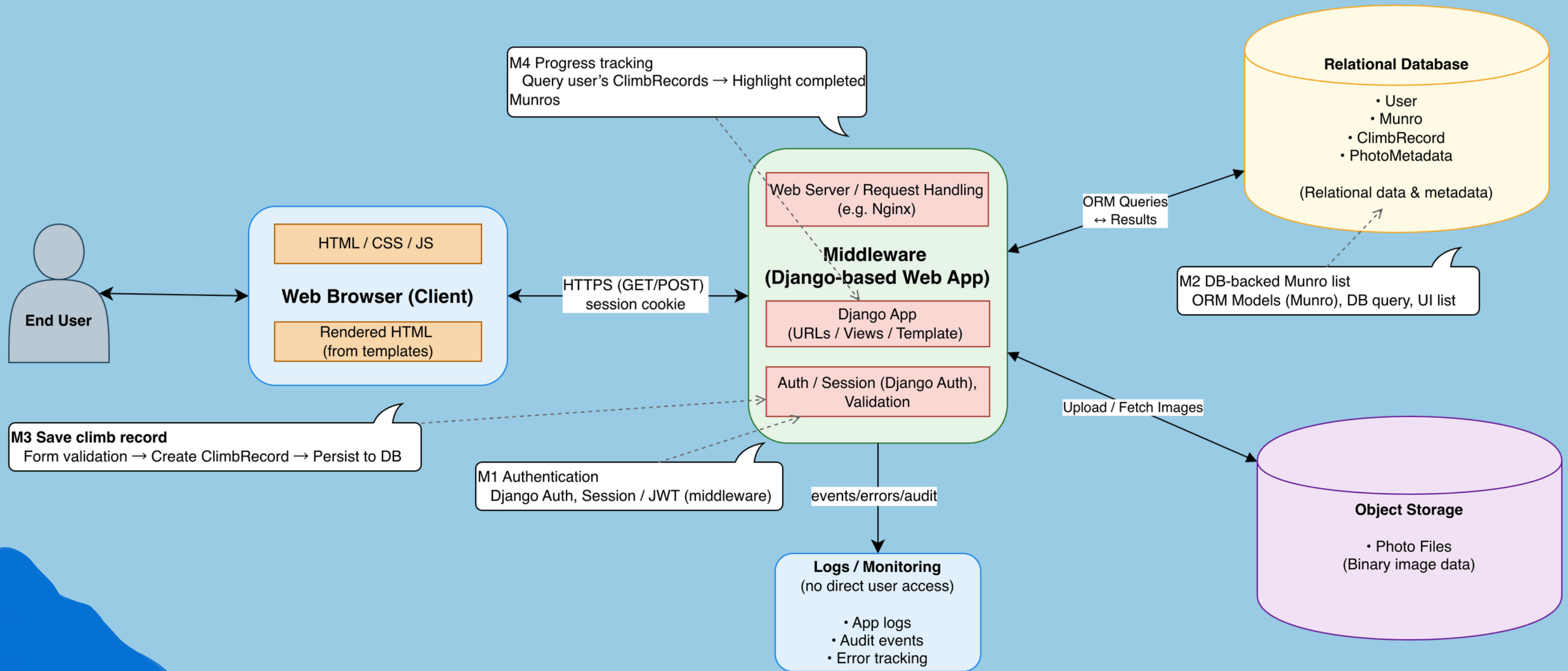
As a user, I want to upload photos for each completed climb record, so that I can visually document my experiences.

### C2 – Advanced Filters

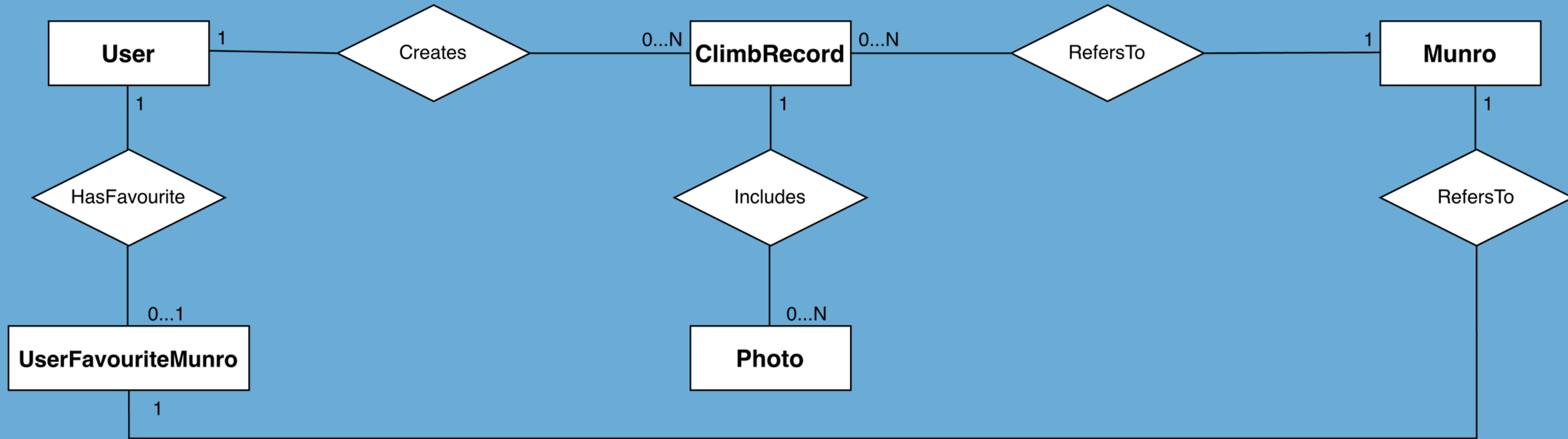
As a user, I want to filter Munros by difficulty or region, so that I can choose climbs that match my ability or location.

# 3.High-level System Architecture

Must requirements mapped to architecture via: M1 Auth/Session, M2 DB-backed Munro model, M3 Persisted ClimbRecord input, M4 Progress tracking and highlighting.



# 4.ER Diagram



## Compressed Chen ER Notes

- Referential integrity:  $\text{ClimbRecord} \rightarrow \text{User} (1)$ ,  $\text{ClimbRecord} \rightarrow \text{Munro} (1)$ ,  $\text{Photo} \rightarrow \text{ClimbRecord} (1)$ ,  $\text{UserFavouriteMunro} \rightarrow \text{User} (1)$ ,  $\text{UserFavouriteMunro} \rightarrow \text{Munro} (1)$ .
- ClimbRecord is an associative entity between User and Munro (M:N resolved).
- Multiple ClimbRecords per (User, Munro) are allowed to support repeat climbs.
- Completion rule: a Munro is completed for a user if  $\geq 1$  ClimbRecord exists.
- Each User has at most one favourite Munro (UserFavouriteMunro.UserID is PK).
- Photo is existence-dependent on ClimbRecord.
- Attributes and types are listed in the tables below.



# 4.ER Diagram

## User Attributes

| Field         | Type         | Constraints/Notes              |
|---------------|--------------|--------------------------------|
| <u>UserID</u> | Integer      | Auto-increment, Unique         |
| UserName      | Varchar(100) | Required, Unique               |
| Email         | Varchar(254) | Required, Unique               |
| PasswordHash  | Varchar(255) | Stored securely (Django auth). |
| CreatedAt     | Datetime     | Auto-set on create             |

## Photo Attributes

| Field                 | Type         | Constraints/Notes                   |
|-----------------------|--------------|-------------------------------------|
| <u>PhotoID</u>        | Integer      | Auto-increment                      |
| RecordID (FK)         | Integer      | Required                            |
| StorageKey / ImageURL | Varchar(512) | Required, Pointer to object storage |
| UploadedAt            | Datetime     | Auto-set on upload                  |
| ContentType           | Varchar(100) | e.g., image/jpeg                    |

## UserFavouriteMunro Attribute

| Field         | Type     | Constraints/Notes                |
|---------------|----------|----------------------------------|
| <u>UserID</u> | Integer  | Required, one favourite per user |
| MunroID (FK)  | Integer  | Required                         |
| SetdAt        | Datetime | Auto-set on create               |

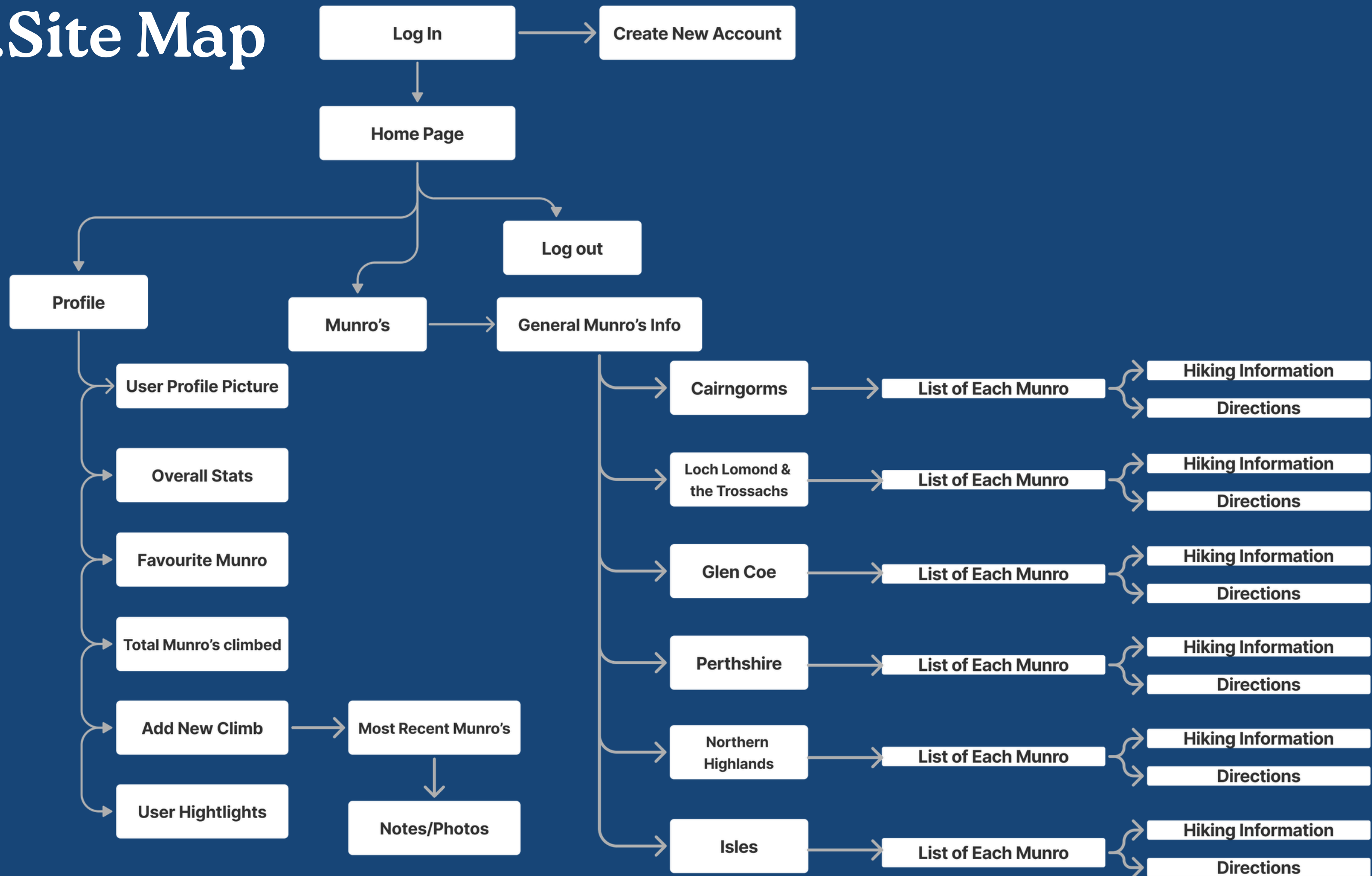
## Munro Attributes

| Field                 | Type         | Constraints/Notes                   |
|-----------------------|--------------|-------------------------------------|
| <u>MunroID</u>        | Integer      | Auto-increment / dataset id         |
| Name                  | Varchar(200) | Required                            |
| Height                | Integer      | Metres                              |
| Location              | Varchar(200) | Required                            |
| Region                | Varchar(100) | Required, Filterable                |
| DifficultyRating      | Integer      | objective difficulty, 1–5           |
| EstimatedTimeHours    | Decimal(4,2) | Optional                            |
| Description           | Text         | Required                            |
| StorageKey / ImageURL | Varchar(512) | Required, Pointer to object storage |
| ContentType           | Varchar(100) | e.g., image/jpeg                    |

## ClimbRecord Attributes

| Field                 | Type     | Constraints/Notes           |
|-----------------------|----------|-----------------------------|
| <u>RecordID</u>       | Integer  | Auto-increment              |
| UserID (FK)           | Integer  | Required                    |
| MunroID (FK)          | Integer  | Required                    |
| ClimbDate             | Date     | Required                    |
| TotalMetersClimbed    | Integer  | Required                    |
| TotalDistance         | Integer  | Required                    |
| CompletionTimeMinutes | Integer  | >=0, required               |
| StarRating            | Integer  | user subjective rating, 1–5 |
| Comments              | Text     | Optional                    |
| CreatedAt             | Datetime | Auto-set on create          |

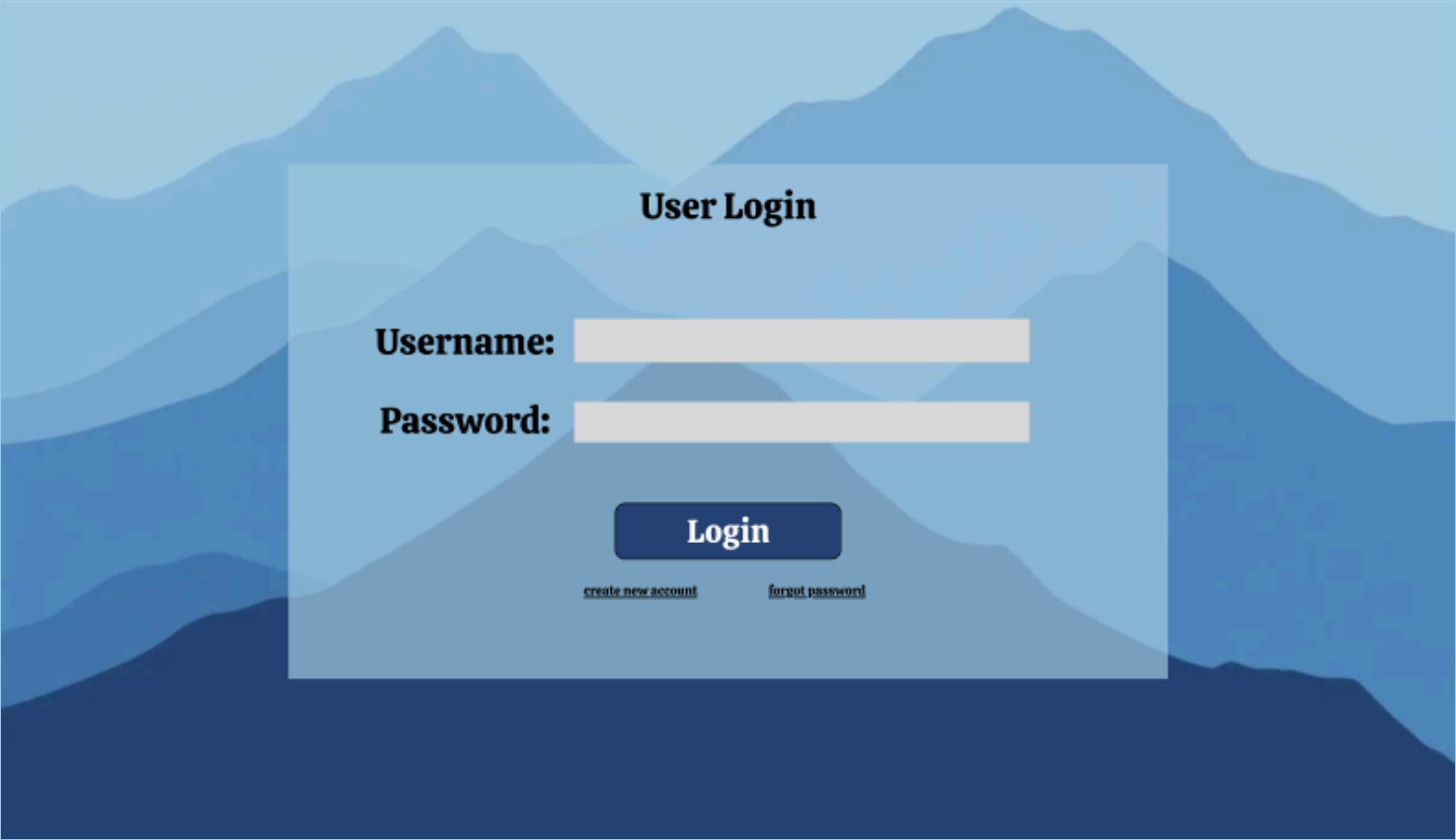
# 5.Site Map



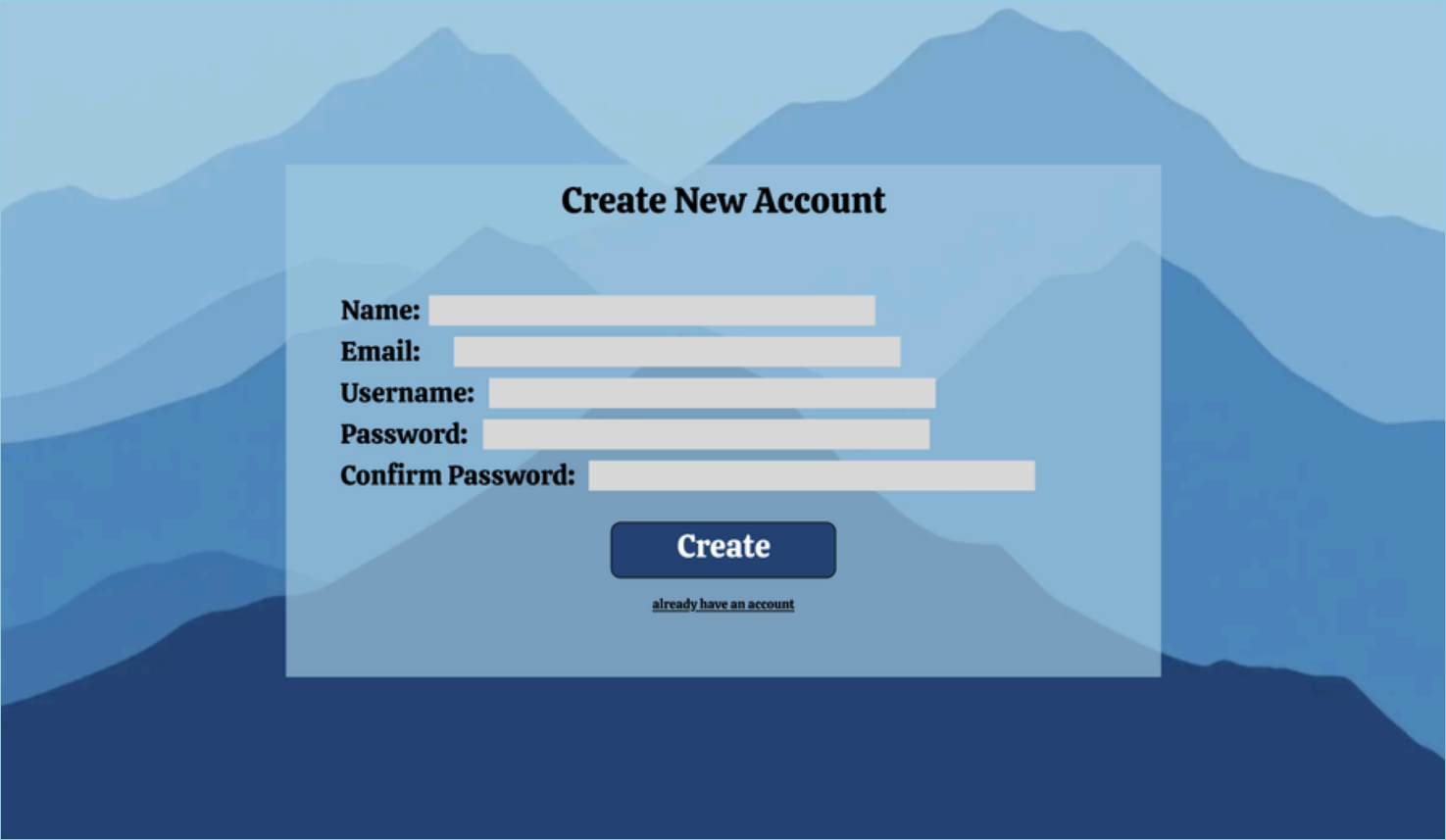
# 6. Wireframes



When accessing the website, the basic homepage will allow the user to log in or create a new account.



User authentication page will prompt user for login details. It will have an option below the login button to create a new account or to reset a password.



Upon clicking “Create New Account,” a new page will appear. It will ask the user for a name, username, email, and password. It will also ask to confirm password (to prevent error). There will be an escape button to head back to main login page (as an emergency exit).



# 6. Wireframes



- Profile
- Munro's
- Logout

Clicking on the hamburger menu will show three accessible link.

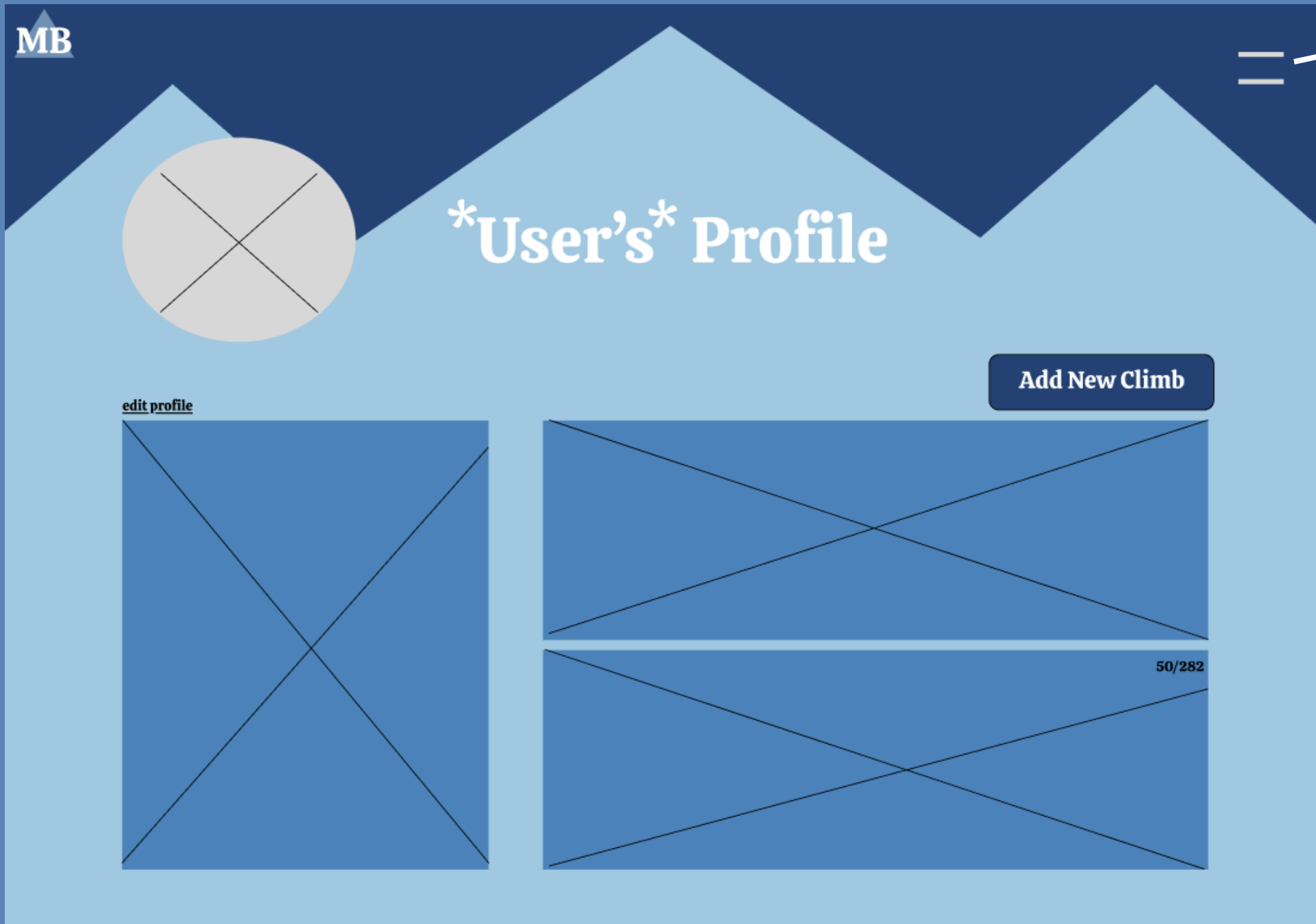
1. Users profile
2. Scotland's Munro's page,
3. Logout

The user will arrive at the landing page after authentication. The landing page will be a high-level view of the site:

1. 'The Munro Bag' title
2. Search bar
3. Hamburger menu on the top right corner.
4. Top 10 highest Munros' in Scotland (with accessible links)
5. Basic Munro information
6. Highlights of the user profile - showcasing the latest Munro climbed.

When clicking on the headers of these three 'boxes', you will be taken to the corresponding page.

# 6. Wireframes



Under Profile, users will be able to upload a profile picture and update their account information, such as first name, email, date of birth, profile picture, etc.

There will be a section on the profile that will focus on overall stats. Users will also be able to update their favourite Munro, total Munro's climbed so far, total overall km, and how long they've been climbing.

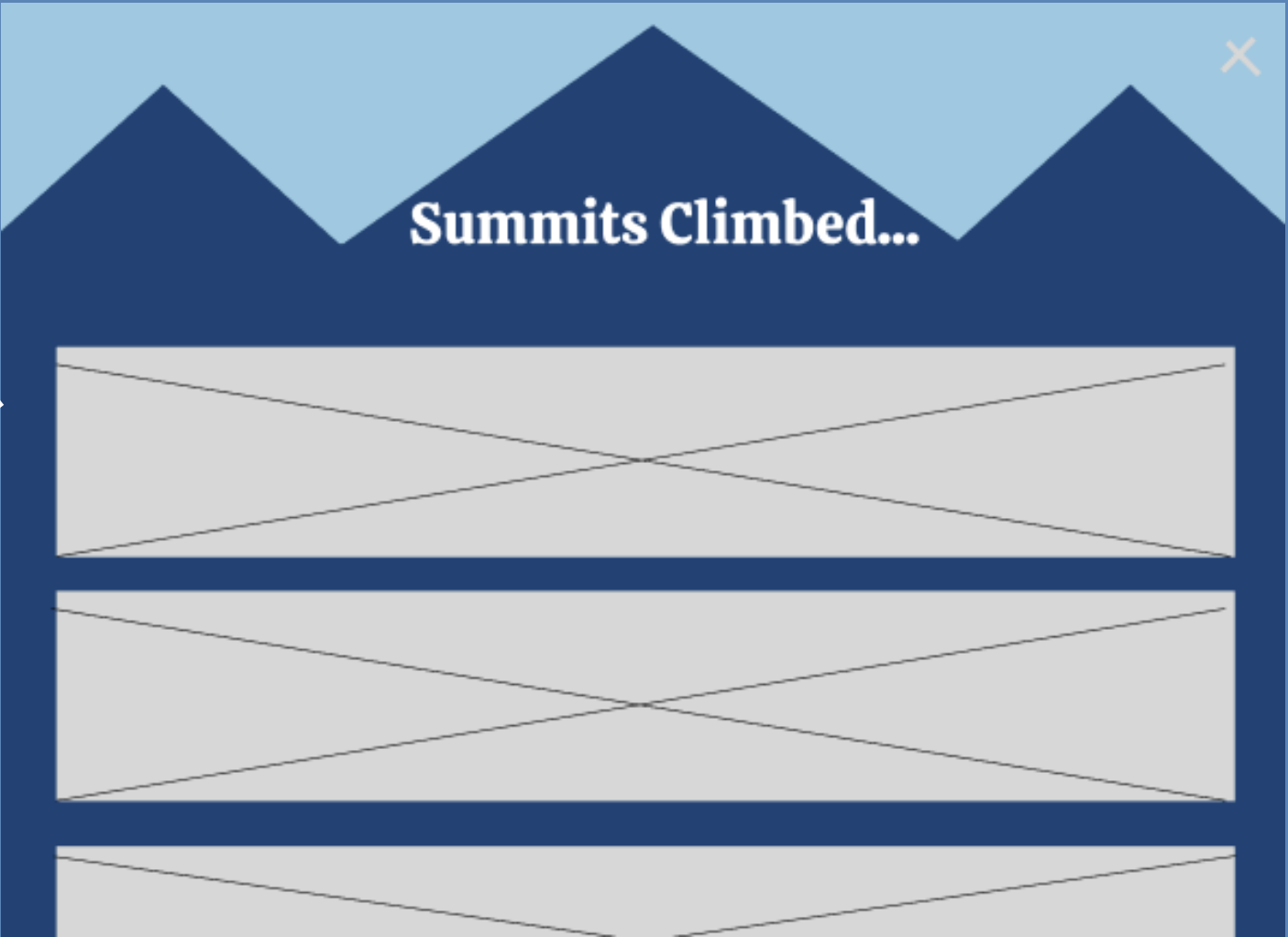
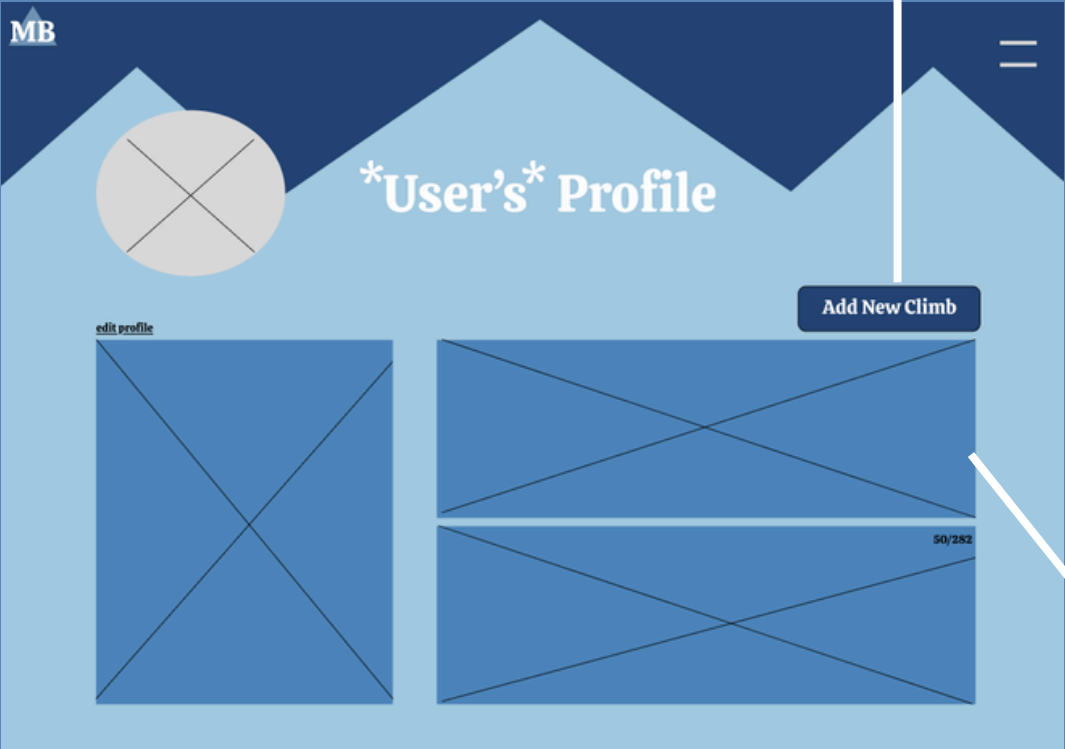
Another area will highlight the most recent Munro completed along with previously completed Munro's out of 282 (total Munro's in Scotland).

Users will be able to select 'Add New Climb' which will be shown in the next slide.

The MB logo on the top left-hand corner acts as an 'exit' to the landing page.

# 6. Wireframes

The user can click ‘Add New Climb’ after completing a Munro. This information will be saved to the database of our users and used to fill out their user profile. All information entered will update their stats. With the submission of this new form, their ‘Most Recent Munro’ section will change to highlight all the information given.



**Congrats, you've reached another summit!**  
Please fill out the following to submit the new climb to your profile

**Munro:**

**Climb date:**  /  /   
format: day/month/year

**Total meters climbed:**  m

**Total distance:**  km

**Total time:**  hr  min

**Rating:** ★★★★★

**Notes:**

**Is this your new favourite?** ☐ yes ☐ no

[Upload images](#)

**Submit**

To see past hikes, notes, and photos, the user can click on the box that contains their ‘Most Recent Munro’. A page will pop up consisting of all past ‘Summits Climbed’ ordered by date.

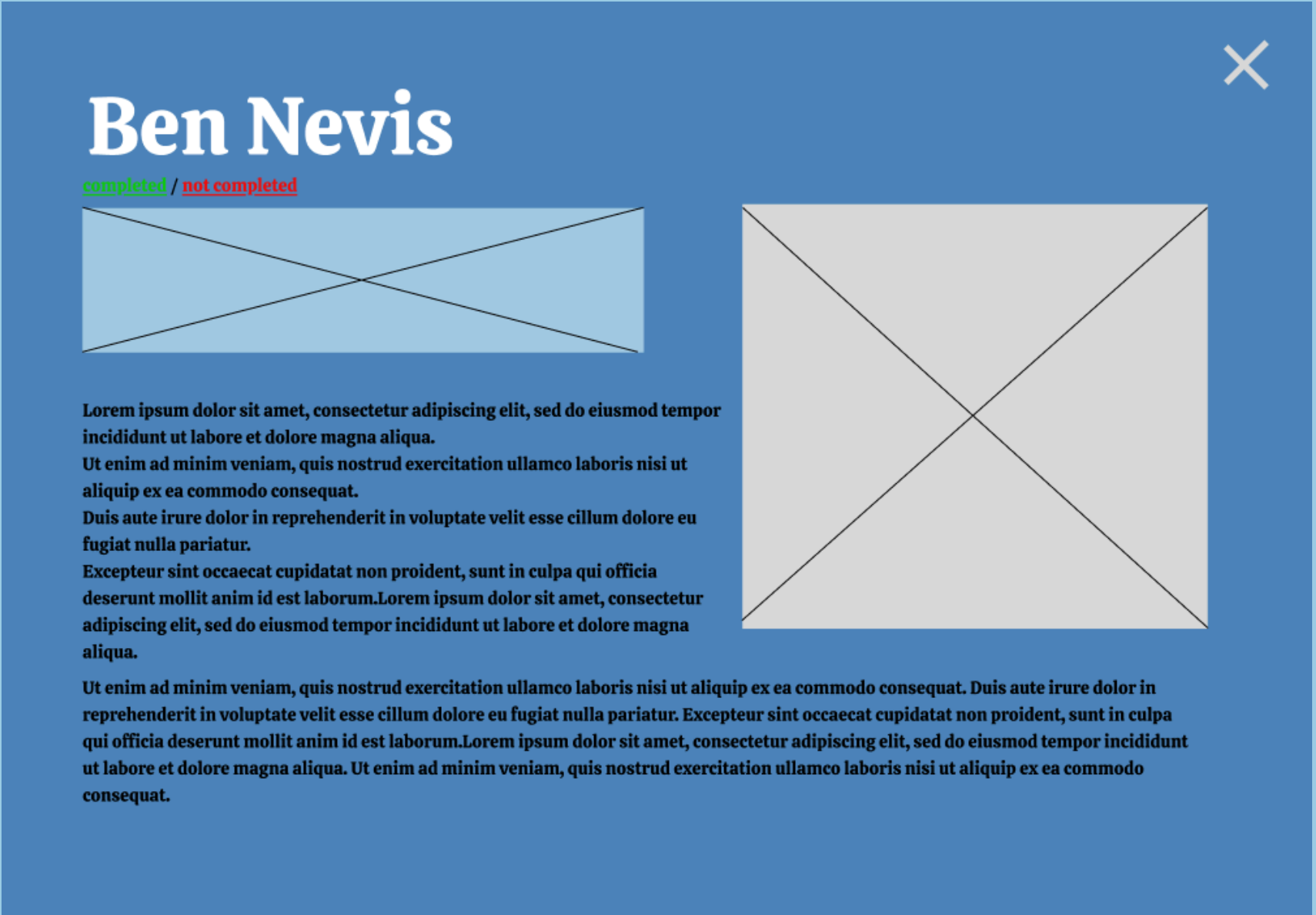
# 6. Wireframes



Once a Munro is selected, the below pop-up page will appear.

The name of the Munro and other information such as hiking information, how to get there, a picture of the Munro, etc.

There will also be a 'completed' or 'not completed' right below the Munro's name. Users will be able to add Munro's to their profile from this page, as well.



On the 'All About Munro's' page, there will be basic information about Munro's and a general search function. A list of all the Munro's divided by region. Each region, users will see hiking information and directions for that specific Munro.

The 'MB' logo in the top left- hand corner will exit to the landing page.



# 7. Accessibility Plan

The five web content accessibility guidelines we plan to follow in our design:

## Colour Contrast 1.4.3

Within our design, we will ensure proper colour contrast. Ensuring text is easily perceivable on our application, through appropriate colour choices, along with the size and weight of text. You will see within our wireframe, that our colour palette consists of a variety of blues. We have adjusted text colour to contrast the deepness/lightness of the blue.

## Link Purpose 2.4.9

The design of our web application will ensure visually distinct link text. Anything that can be 'clicked' within our web application, will be underlined (as shown in our wireframe) and will change colour when hovered over. This will ensure perceivability for our users.

## Headings & Labels 2.4.6

Clear directional labels and headings are an essential part in a cohesive and understandable application. To signal our homepage, it will have clear, large font in the centre of the page, with our 'main' mountain background. Our side pages will have a different mountainous background and the title will not be centred in the page.

## Focus Order 2.4.3

In our top righthand corner, we will have a hamburger menu that allows you to switch between pages. When switching pages, the tab flow will change. Always ensuring the following order: Home → Profile → Munro's → Logout. You can see how this is implemented within our wireframes.

## Input Assistance 3.3

Alerting users to errors and/or trying to prevent errors from occurring is an important part of design. You'll see this in our user authentication page, where we will ensure the user to type their new password twice. We also have an 'emergency exit' that can take a user back to the login. For our 'new climb' form, we will put in some safeguards as well. The Munro will be selectable from a list, ensuring correct spelling/linkage to our database. There will also be a minimum and maximum climbing limit, based on Munro requirements.

# 8. Appendix

## Member Contribution

Nicole Cohen

3121217C

Contribution = 33.33%

Concept

Wireframes

Accessibility

Makha Oyanon Taruk

3162543O

Contribution = 33.33%

User Stories

Overview

Sitemap

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Contribution = 33.33%

User Stories

System Architecture

ER Diagram

## AI Usage Statement

1. We declare that we have used GenAI for copy-editing / improving clarity of language

## Sources Used

World Wide Web Consortium. (2023). Web Content Accessibility Guidelines (WCAG) 2.2: Quick reference.

<https://www.w3.org/WAI/WCAG22/quickref/#principle1>